Solution Concentrations, Stoichiometry and Titrations

DANGER! CAUSES SEVERE BURNS MAY BE FATAL If SWALLOWED	
Specifications:	
CH_3COOH M.W., g/mol	60.0
Assay %, W/W	99.50
Formic Acid, ppm (max)	1000
Total Chloride, ppm, (max)	15
Heavy Metal as Pb, ppm, (max)	5
Freezing Point (min), °C	15.6
Initial Boiling Point (min), °C	116.5
Specific Gravity, g/mL (27°C/27°C)	1.0496
Colour (APHA)	10
Iron, ppm, (max)	2
Solubility in Water	Soluble

1. You have been given a larger beaker, a 200 mL graduated cylinder and a sample of solid $FeSO_{4(s)}$. List the specific the procedural quantitative steps required to prepare a 150 mL solution of 0.22 M $FeSO_4$.

2. a) What volume of 12.0 M HCl is used to produce 225 mL of 0.14 M HCl?
   b) What volume of water must be added to 28 mL of 4.6 M NaCl solution to dilute it by 25%?

3. Use the adjacent information from the label on a bottle of stock (glacial) acetic acid to determine its concentration.

4. Consider the following reaction: $Bi_2S_{3(s)} + HCl_{(aq)} \rightarrow BiCl_{3(s)} + H_2S_{(aq)}$. When excess $Bi_2S_{3(s)}$ is added to 125 mL of $HCl_{(aq)}$, 3.8 g of $BiCl_{3(s)}$ was predicted to be formed. What was the concentration of the original $H_2S_{(aq)}$ solution formed as well?

5. What volume of 1.8 M $Ba(OH)_2$ is required to completely neutralize 72 mL of 1.4 M citric acid, $H_3C_6H_5O_7$ (a triprotic acid)?

6. a) Identify similarities between the titrant and the primary standard used in titration.
   b) Identify differences between the titrant and the primary standard used in titration.

7. a) Explain how the equivalence point of an acid/base titration is determined mathematically.
   b) Explain how the equivalence point of an acid/base titration is determined visually.